Rupinder Kaur

Doctor of Philosophy
in Computer Science
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Education

Degree	Institute/Board	CGPA/Percentage	Year
Doctor of Philosophy	Indian Institute of Technology Ropar	8.09	2023-Present
Master of Science in Information Technology	G.G.D.S.D. College, Chandigarh	79.19%	2018-2020
Bachelor of Computer Science	D.A.V. College, Jalandhar	60.4%	2014-2017

Projects

- **Derivations and Applications of Computer Science in Mathematics** *2024*
  Coursework: Mathematical Foundation of Computer Science
  - This project is based on key mathematical concepts in computer science and machine learning, specifically Least Squares Methods (Linear and Non-Linear), Principal Component Analysis (PCA) and Kernel PCA, Singular Value Decomposition (SVD), and Kernel SVD. The project explained how these techniques work, their derivations, and their practical applications in machine learning and data analysis.
- **GDP Analysis using Machine Learning Techniques** *2024*
  Coursework: Machine Learning
  - In this project, we analyzed the relationship between socio-economic factors such as life expectancy, education, tourism, unemployment, and environmental factors affecting the GDP of different countries using a dataset of 204 countries. I performed data preprocessing techniques including handling missing values, normalization, feature selection, and PCA for dimensionality reduction. Through this project, I learned how to preprocess real-world datasets, reduce data dimensions, and evaluate and compare different unsupervised machine learning models.
- **Coffee Shop Management Website** *2020*
  Minor Project of Master's Degree
  - Developed a web-based coffee shop management system using Angular and PHP where customers can place and cancel orders with automated bill generation. The system also monitored daily sales and analyzed monthly income to improve business strategy.

Publications

1. **Rupinder Kaur**, Shahbaz Ahmad Khanday, Simrandeep Singh, Uday Thakur, Ashish Verma, and Mukesh Saini. "TATVA: Turmeric Adulteration Detection using Thermal Video Analysis." In *MMFood '25: Proceedings of the 1st International Workshop on Multi-modal Food Computing*, ACM, 2025. https://doi.org/10.1145/3746264.376049

2. Shreya Bansal, Ruchi Bhatt, Amanpreet Chander, **Rupinder Kaur**, Malya Singh, Mohan Kankanhalli, Abdulmotaleb El Saddik, and Mukesh Saini. "GroMo25: ACM Multimedia 2025 Grand Challenge for Plant Growth Modeling with Multiview Images." In *Proceedings of the 33rd ACM International Conference on Multimedia*, ACM, 2025. https://doi.org/10.1145/3746027.376209

3. **Rupinder Kaur**, Simrandeep Singh, and Mukesh Saini. "Machine Learning Based Approach to Detect Adulteration in Turmeric Using RGB and Thermal Images." In *Agricultural-Centric Computation (ICA 2024)*, Springer, 2025. https://doi.org/10.1007/978-3-031-74440-2_16

Achievements

- **Best Women Innovation Award**, Zentej S2 Hackathon 2025 *2025*
- **UGC NET (December 2022)**, Score: 162/300 *2022*
- **UGC NET (June 2023)**, Score: 158/300 *2023*
- **GATE 2022**, GATE Score: 240 *2022*
- **GATE 2023**, GATE Score: 232 *2023*

Technical Skills

- **Programming Languages**: C/C++, PHP, Python
- **Tools Libraries**: TensorFlow, Pytorch

Key Courses Taken

- **CSE & Maths**: Deep Learning for Computer Vision, Machine Learning, Mathematical Foundation of Computer Science, Multimedia Surveillance Systems, Research Methodology in Computer Science
- **Others**: Introduction to Agriculture Cyber Physical Systems, Internet of Things, Administration and Management of Engineering Education

Additional Details

- **Volunteer**, 10th International Conference on Computer Vision & Image Processing (CVIP) *2025*
- **6-Month Industrial Training**, Industrial Training in Web Designing and Development using Angular and PHP *2020*
- **National Cadet Corps (NCC)**, Awarded NCC 'A' Certificate *2014*